



POWER BI MAJOR CAPSTONE PROJECT
END TO END
POWER BI PROJECT GUIDE

PROJECT TITLE

SUPERSTORE SALES & PROFIT
ANALYTICS DASHBOARD

PROJECT CATEGORY

Sales Analytics
Profit Analytics
Business Intelligence

INSTITUTE NAME

Pallavi Engineering College

TRAINER / GUIDE

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CREATED BY

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TOOLS USED

Power BI Desktop Power Query, DAX &
Power BI
Service

SUBMISSION DATE

[09/06/2026]

Prepared as part of the Power BI Masterclass Program to demonstrate skills in Data Analysis, Data Modeling, DAX Development, Dashboard Design, UI/UX, Storytelling, and Business Intelligence Reporting.

PROJECT OVERVIEW

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PROJECT OBJECTIVE

Develop a comprehensive Power BI dashboard to analyze sales, profit, customers, products, and regional performance for a retail business

Participants are expected to:

- ✓ Analyze sales performance
- ✓ Evaluate profitability
- ✓ Understand customer behavior
- ✓ Assess regional performance
- ✓ Generate business insights
- ✓ Present recommendations

PROJECT GOAL

Transform transactional retail data into meaningful business insights that support decision-making, performance monitoring, and business growth.

BUSINESS PROBLEM

Retail management requires a clear understanding of sales performance, profitability, customer behavior, and product contribution to make better business decisions.

The objective is to identify:

- Which products generate the highest sales?
- Which regions contribute the most profit?
- Which customer segments drive business growth?

- How discounts impact profit?
- Which products require business attention?

BUSINESS QUESTIONS

Participants must answer:

1. What is total sales revenue?
2. What is total profit?
3. Which products perform best?
4. Which regions generate maximum revenue?
5. Which customer segments contribute most?
6. How do discounts affect profit?
7. What are the sales and profit trends over time?

DATASET INFORMATION

DATASET NAME : Superstore Dataset

DATASET DETAILS : Approximate Records 9,994+

Columns : 19+

EXPECTED ANALYSIS AREAS

Sales Analytics

Profit Analytics

Customer Analytics

Product Analytics

Regional Analytics

Superstore Sales & Profit Analysis POWER QUERY TRANSFORMATIONS

DATACLEANINGACTIVITIES

✓ Removed Duplicate Transactional Records

✓ Validated Sales and Profit Numerical Data

- ✓ Handled Missing Values and Blank Categories
- ✓ Corrected Data Types (Dates, Currency, Whole Numbers)
- ✓ Standardized Regional and Segment Categories

DATAPREPARATIONACTIVITIES

- ✓ Renamed Columns for Clean Visualization Formatting
- ✓ Created Custom Columns for Business Logic Slicing
- ✓ Replaced Values for Incomplete or Blank Product Fields
- ✓ Split Columns (e.g., Extracting Product Type from SKU Strings)
- ✓ Merged Columns (e.g., Unifying Customer Names or Complete Addresses)
- ✓ Removed Unnecessary Operational Columns to Optimize Performance

SCREENSHOT 3 Power Query Editor Screenshot

Queries [3] < X ✓ f_x Table.TransformColumnTypes(#"Promoted Headers",({{"Product ID", type text}, {"Product Name", type text}}))

	Product ID	Product Name
	Valid 100%	Valid 100%
	Error 0%	Error 0%
	Empty 0%	Empty 0%
1	FUR-BO-10001798	Bush Somerset Collection Bookcase
2	FUR-CH-10000454	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back
3	OFF-LA-10000240	Self-Adhesive Address Labels for Typewriters by Universal
4	FUR-TA-10000577	Bretford CR4500 Series Slim Rectangular Table
5	OFF-ST-10000760	Eldon Fold 'N Roll Cart System
6	FUR-FU-10001487	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood
7	OFF-AR-10002833	Newell 322
8	TECH-PH-10002275	Mitel 5320 IP Phone VoIP phone
9	OFF-BI-10003910	DXL Angle-View Binders with Locking Rings by Samsill
10	OFF-AP-10002892	Belkin F5C200VTEL 6 Outlet Surge
11	FUR-TA-10001539	Chromcraft Rectangular Conference Tables
12	TECH-PH-10002033	Konftel 250 Conference phone - Charcoal black
13	OFF-PA-10002365	Xerox 1967
14	OFF-BI-10003656	Fellowes PB200 Plastic Comb Binding Machine
15	OFF-AP-10002311	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HE...
16	OFF-BI-10000756	Stonex DuraTech Recycled Plastic Frosted Binders
17	OFF-ST-10004186	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D
18	OFF-ST-10000107	Fellowes Super Stor/Drawer
19	OFF-AR-10003056	Newell 341
20	TECH-PH-10001949	Cisco SPA 501G IP Phone
21	OFF-BI-10002215	Wilson Jones Hanging View Binder, White, 1"
22	OFF-AR-10000246	Newell 318
23	OFF-AP-10001492	Acco Six-Outlet Power Strip, 4' Cord Length
24	FUR-CH-10002774	Global Deluxe Stacking Chair, Gray
25	OFF-BI-10001634	Wilson Jones Active Use Binders

Query Settings

PROPERTIES

Name

Product_details

All Properties

APPLIED STEPS

Source

Navigation

Changed column type

Promoted Headers

Changed Type

Queries [3]

Abbreviations

Product_details

Data

Table.TransformColumnTypes(#"Promoted Headers",{{"Sub-Category", type text}, {"Sub-Category code", type text}, {"Category", type text}, {"Category code", type text}})

	Sub-Category	Sub-Category code	Category	Category code
	Valid 100%	Valid 100%	Valid 18%	Valid 18%
	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 82%	Empty 82%
1	Bookcases	BO	Furniture	FUR
2	Chairs	CH	Office Supplies	OFF
3	Labels	LA	Technology	TECH
4	Tables	TA		null
5	Storage	ST		null
6	Furnishings	FU		null
7	Art	AR		null
8	Phones	PH		null
9	Binders	BI		null
10	Appliances	AP		null
11	Paper	PA		null
12	Accessories	AC		null
13	Envelopes	EN		null
14	Fasteners	FA		null
15	Supplies	SU		null
16	Machines	MA		null
17	Copiers	CO		null

Query Settings

PROPERTIES

Name

Abbreviations

All Properties

APPLIED STEPS

Source

Navigation

Changed column type

Promoted Headers

Changed Type

Queries [3]

Abbreviations

Product_details

Data

Table.TransformColumnTypes(#"Promoted headers",{{"Row ID", Int64.Type}, {"Order ID", type text}, {"Order Date", type date}, {"Ship Date", type date}, {"Ship Mode", type text}, {"Customer ID", type text}, {"Customer Name", type text}})

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
1	1	CA-2016-152156	08-11-2016	11-11-2016	Second Class	CG-12520	Claire Guter
2	2	CA-2016-152156	08-11-2016	11-11-2016	Second Class	CG-12520	Claire Guter
3	3	CA-2016-138688	12-06-2016	16-06-2016	Second Class	DV-13045	Darrin Van Hous
4	4	US-2015-108966	11-10-2015	18-10-2015	Standard Class	SO-20335	Sean O'Donnell
5	5	US-2015-108966	11-10-2015	18-10-2015	Standard Class	SO-20335	Sean O'Donnell
6	6	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffr
7	7	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffr
8	8	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffr
9	9	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffr
10	10	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffr
11	11	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffr
12	12	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffr
13	13	CA-2017-114412	15-04-2017	20-04-2017	Standard Class	AA-10480	Andrew Allen
14	14	CA-2016-161389	05-12-2016	10-12-2016	Standard Class	IM-15070	Irene Maddox
15	15	US-2015-118983	22-11-2015	26-11-2015	Standard Class	HP-14815	Harold Pawla
16	16	US-2015-118983	22-11-2015	26-11-2015	Standard Class	HP-14815	Harold Pawla
17	17	CA-2014-105893	11-11-2014	18-11-2014	Standard Class	PK-19075	Pete Kriz
18	18	CA-2014-167164	13-05-2014	15-05-2014	Second Class	AG-10270	Alejandro Grc
19	19	CA-2014-143336	27-08-2014	01-09-2014	Second Class	ZD-21925	Zuschuss Don
20	20	CA-2014-143336	27-08-2014	01-09-2014	Second Class	ZD-21925	Zuschuss Don
21	21	CA-2014-143336	27-08-2014	01-09-2014	Second Class	ZD-21925	Zuschuss Don
22	22	CA-2016-137330	09-12-2016	13-12-2016	Standard Class	KB-16585	Ken Black
23	23	CA-2016-137330	09-12-2016	13-12-2016	Standard Class	KB-16585	Ken Black
24	24	US-2017-156909	16-07-2017	18-07-2017	Second Class	SF-20065	Sandra Flana
25	25	CA-2015-160333	25-08-2015	30-08-2015	Standard Class	FB-12030	Frank B...

Query Settings

PROPERTIES

Name

Data

All Properties

APPLIED STEPS

Source

Navigation

Promoted headers

Changed column type

DATA MODELING

Star Schema

Purpose:

FACT TABLE

Employee_Fact

DIMENSION TABLES

Abbreviations

Data

Product details

RELATIONSHIPS

One-to-Many Relationships

MODEL BENEFITS

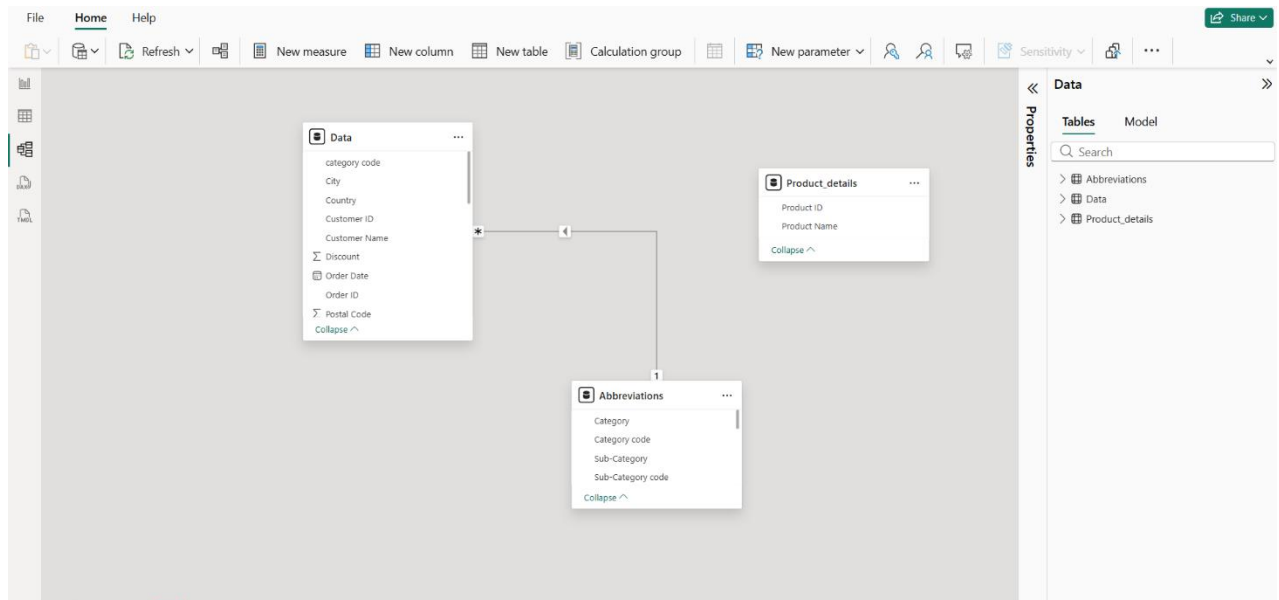
Improved Performance

Easy DAX Development

Better Scalability

SCREENSHOT

Data Model Screenshot



DAX MEASURES USED

Total Sales

Total Sales = SUM(Data[Sales])

PURPOSE : Calculate overall sales.

Total Profit

Total Profit = SUM(Data[Profit])

PURPOSE : Calculate Overall profit

Total Orders

Total Orders = COUNT(Data[Order ID])

PURPOSE : Calculate Total count of order

Top Performance Region

TopPerformanceRegion=CALCULATE(SELECTEDVALUE(Data[Region]),TOPN(1,ALL(Data[Region]),[Total Sales],DESC))

PURPOSE : To check the top performance region ex: east

Avg Order Value

Avg Order Value = DIVIDE([Total Sales],[Total Orders],0)

PURPOSE : to calculate the Average of Orders

Profit Margin %

Profit Margin % = DIVIDE([Total Profit],[Total Sales],0)*100

PURPOSE : Measures the overall Profitability

VISUALIZATIONS USED

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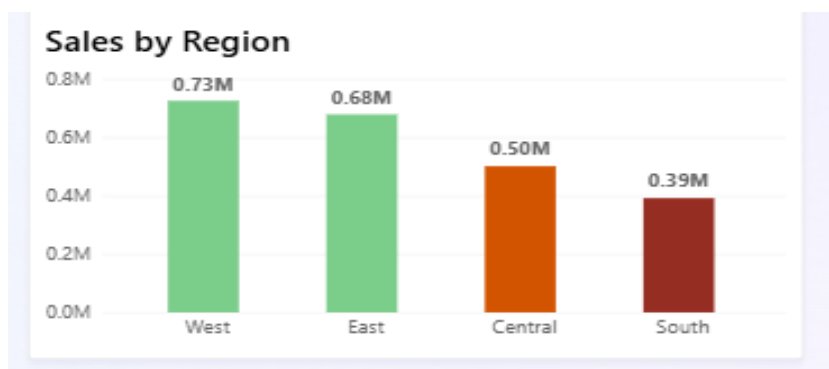
Executive Overview

KPI CARDS: Display key business metrics.

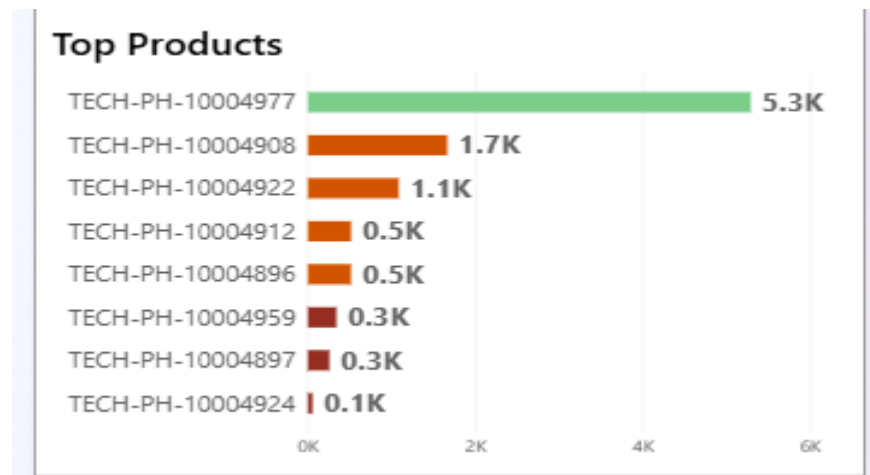
SCREENSHOT



BAR CHART: Shows the Total Sales in Region.

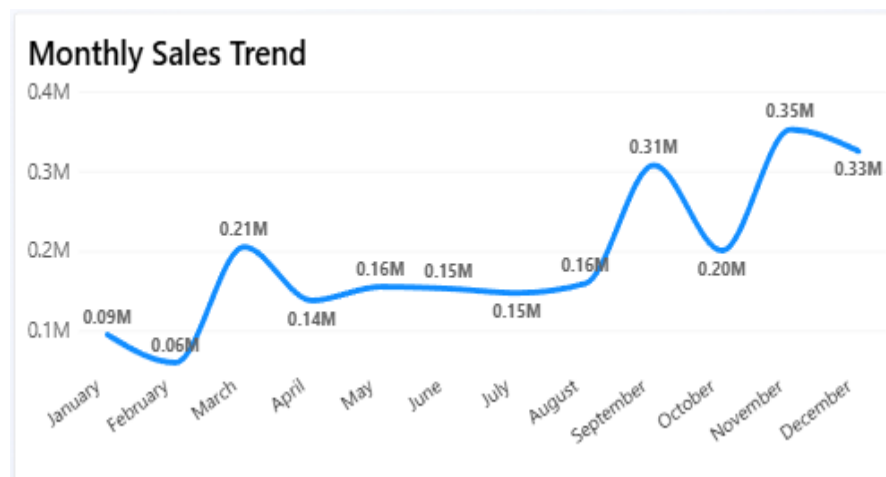


CLUSTERED BAR CHART : Shows the top products



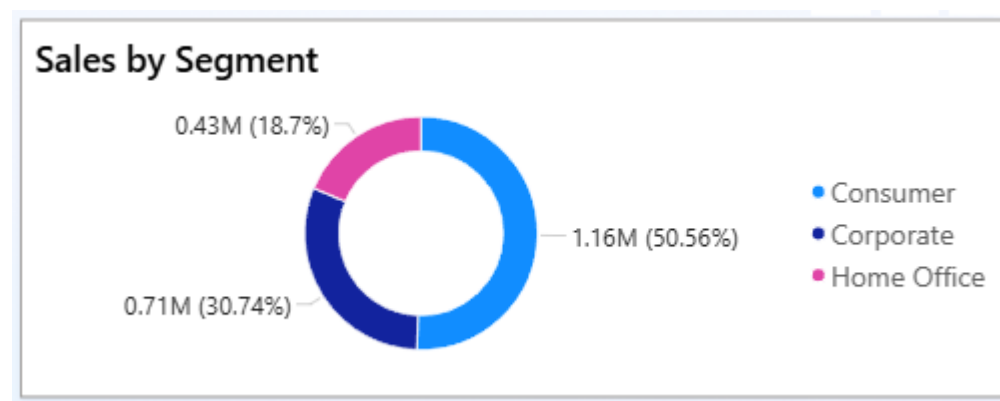
SCREENSHOT

LINE CHART: Show sale trends over time.



SCREENSHOT

DONUT CHART : Shows the Sales according to segments

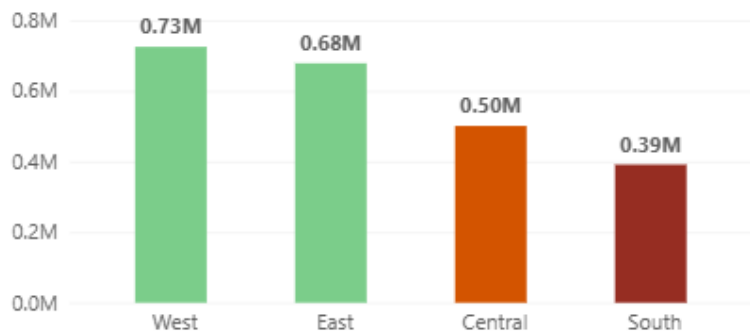


SCREENSHOT

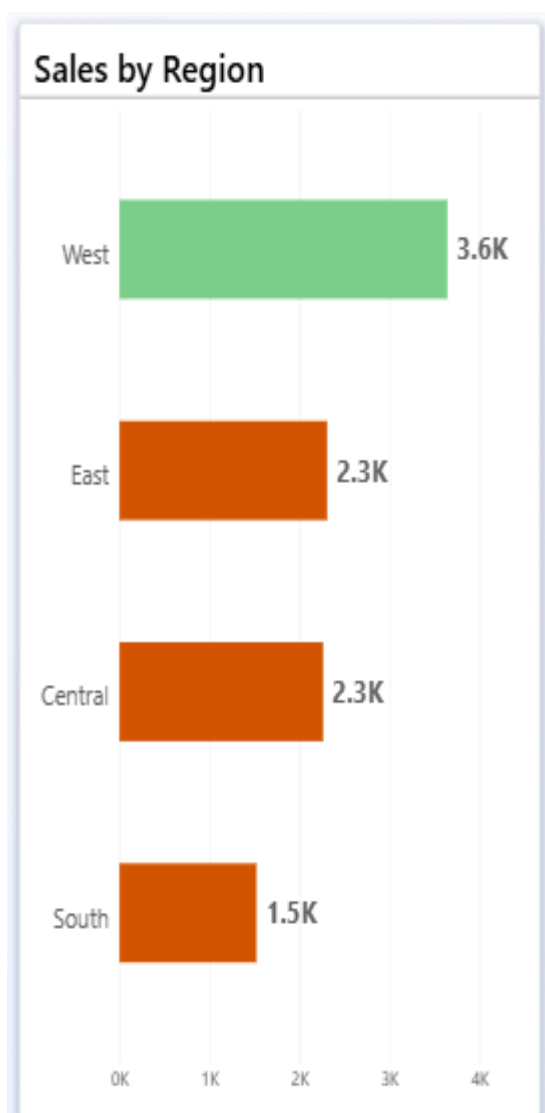
Sales Analytics

COLUMN CHART : Shows the sales according to Category

Sales by Region



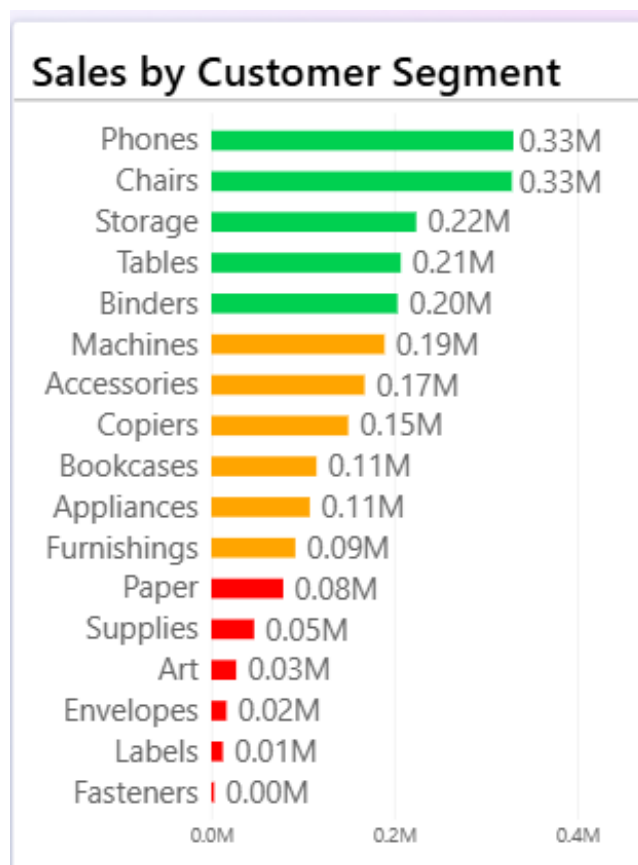
CLUSTERED BAR CHART :Shows the Sales according to Region



LINE CHART : Shows the Sales trend over time



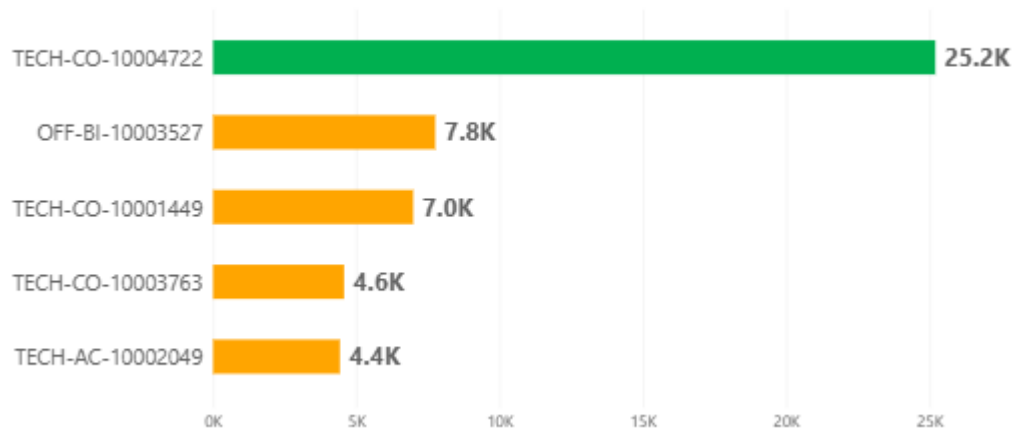
CLUSTERED BAR CHART :Shows the sales by the customer segment



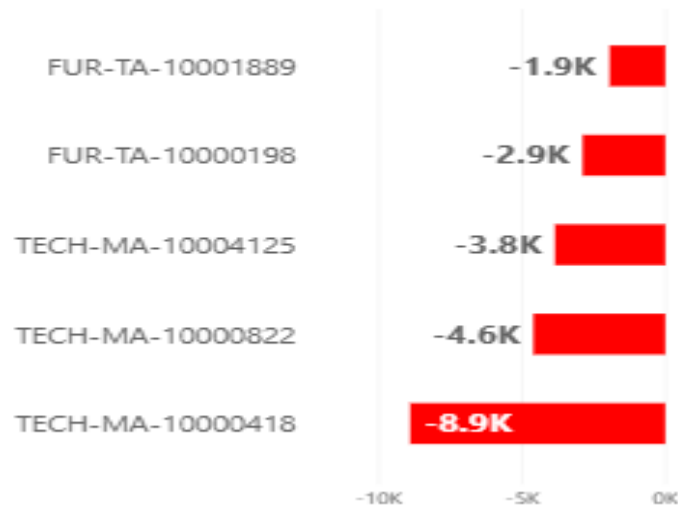
Profit Analytics

CLUSTERED BAR CHART : Shows the Top and Bottom profit Products

Top Profit Products

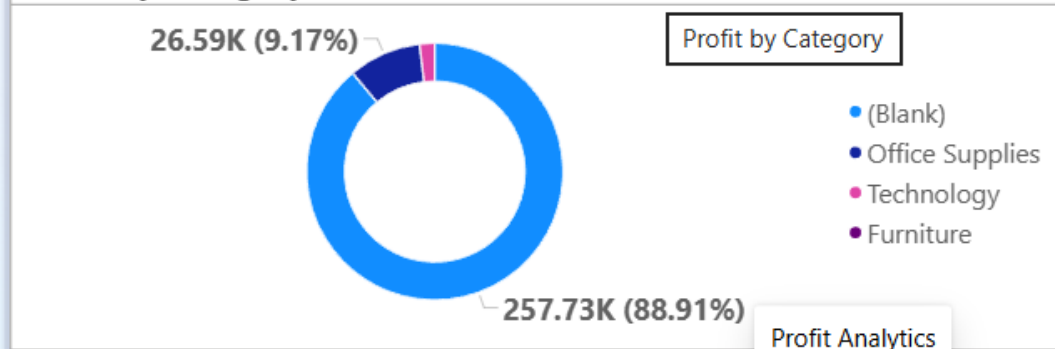


Bottom Profit Products



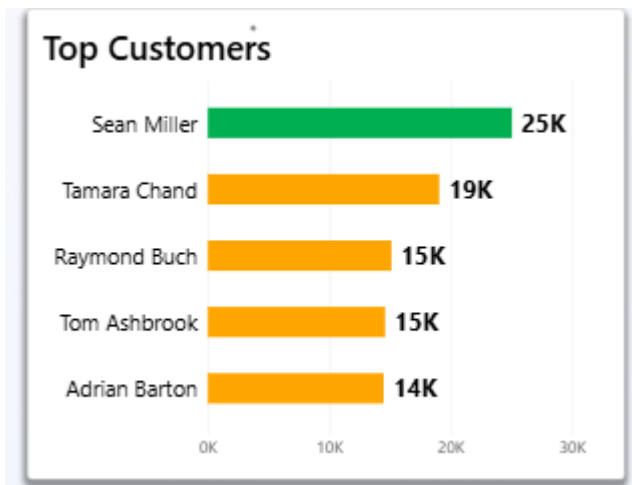
DONUT CHART : Shows the Profit according to Category

Profit by Category

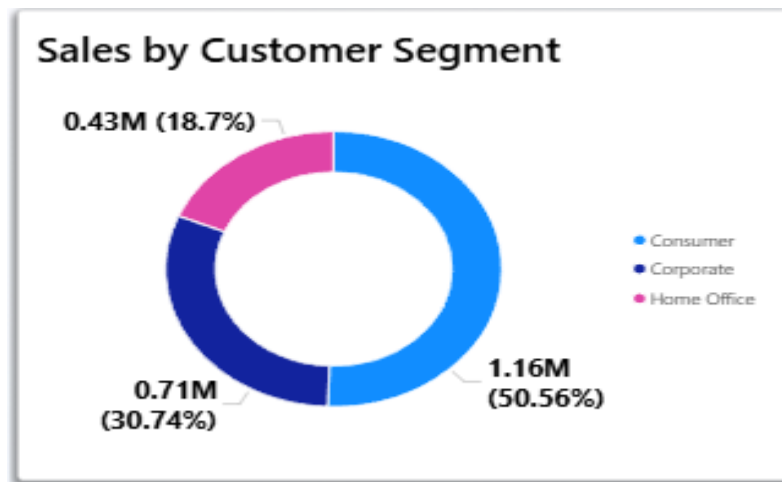


Customer Analytics

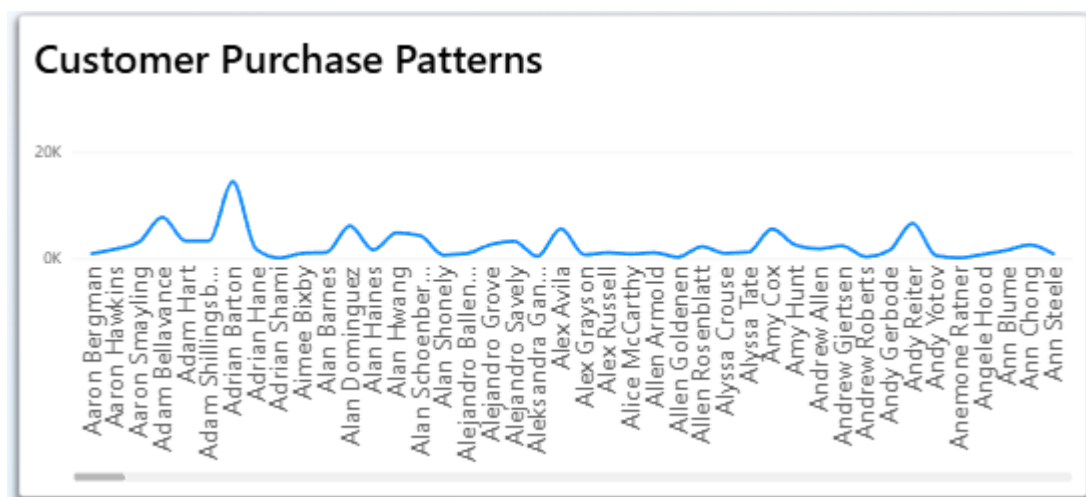
CLUSTERED BAR CHART :Shows the Top Customers



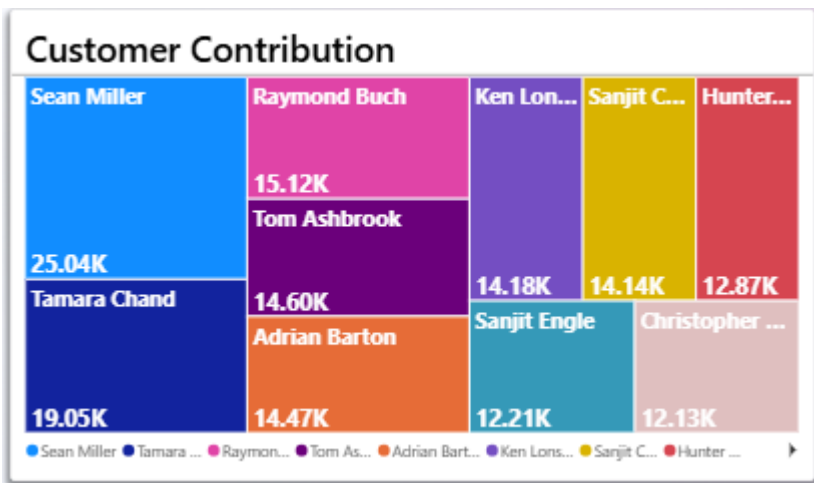
DONUT CHART : Shows the Sales According To Customer Segment



LINE CHART: Analyze Customer Purchase Patters.



TREE MAP : Show the customer contribution

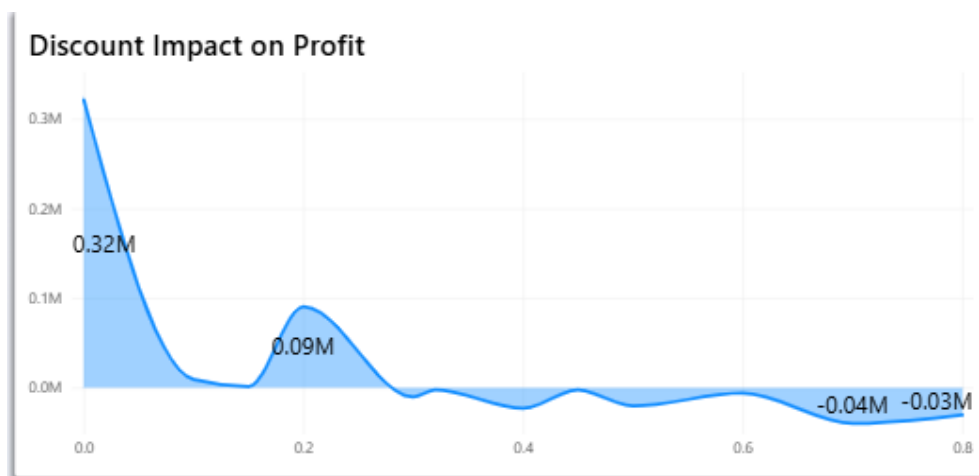


MAP : Shows The Sales according to customer id and Region



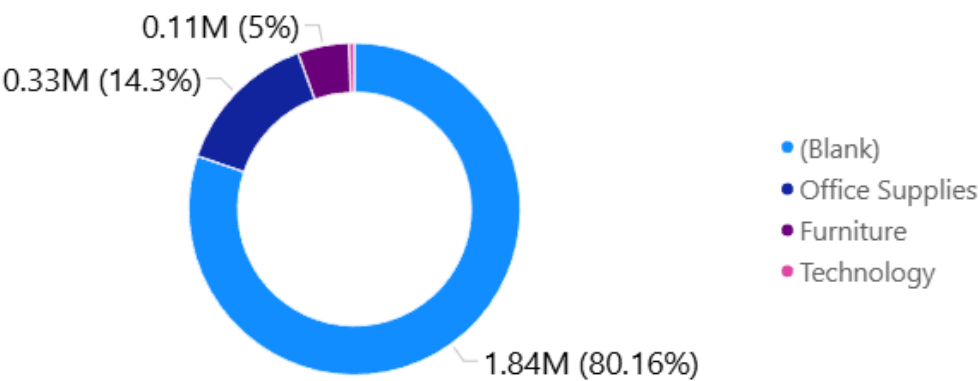
Product Analytics

SCATTER CHART: Discount vs Product

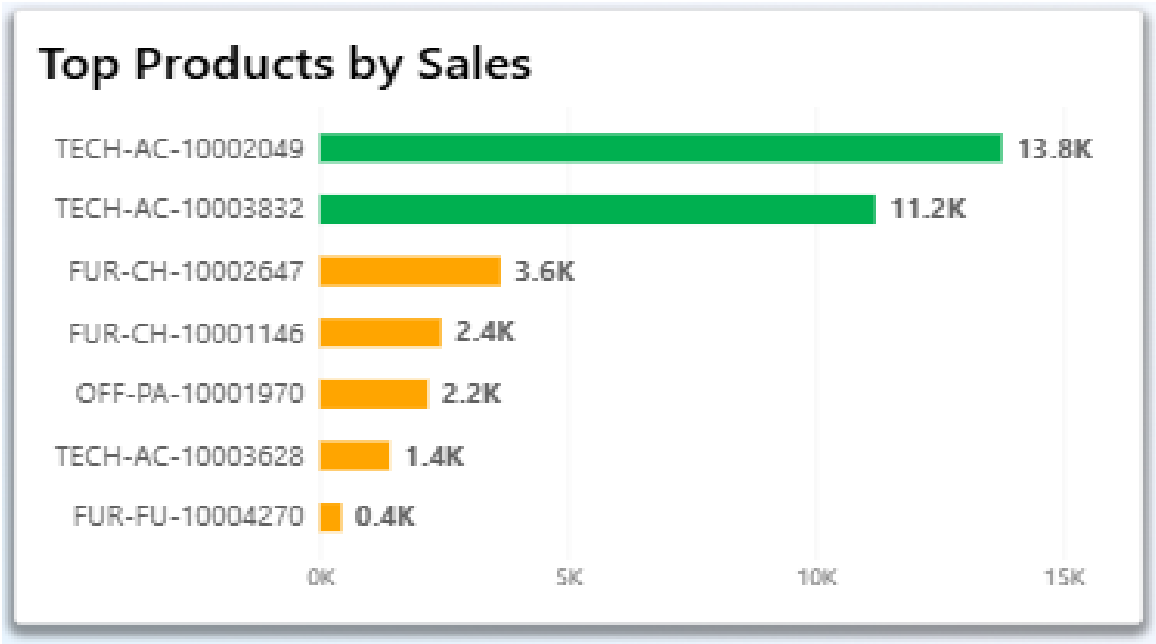


DONUT CHART : Shows the Category according to sales

Category Contribution



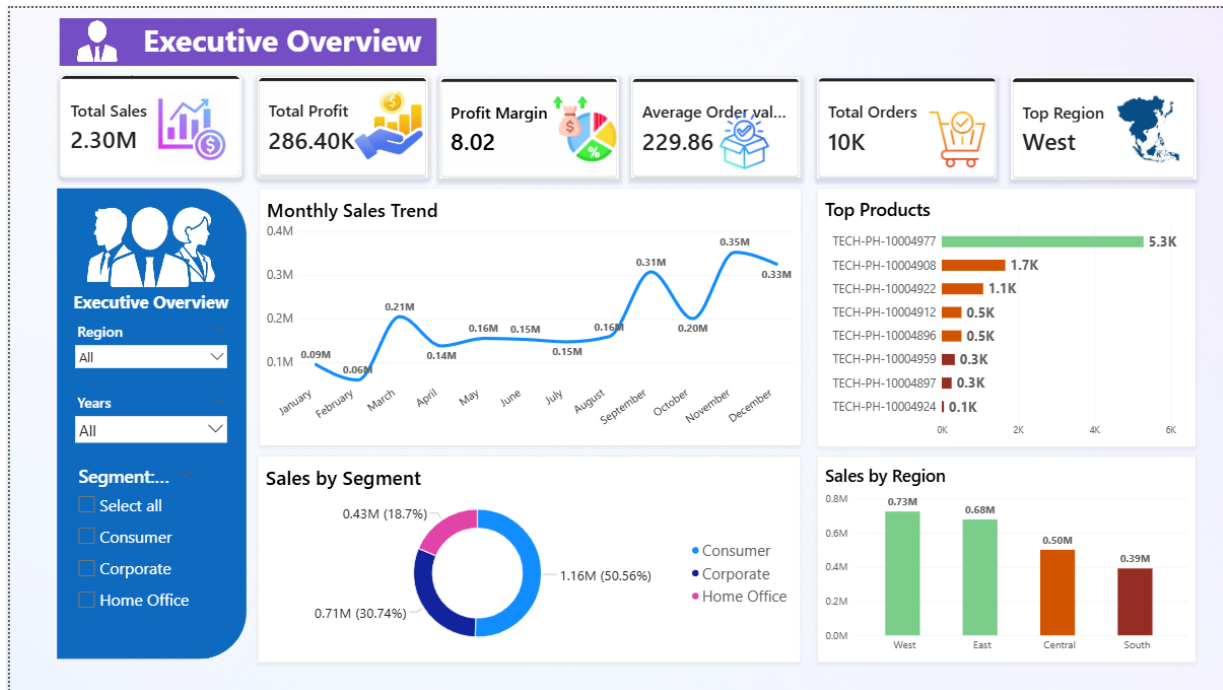
CLUSTERED BAR CHART : Shows the Top profit by Sales



DASHBOARD SCREENSHOTS

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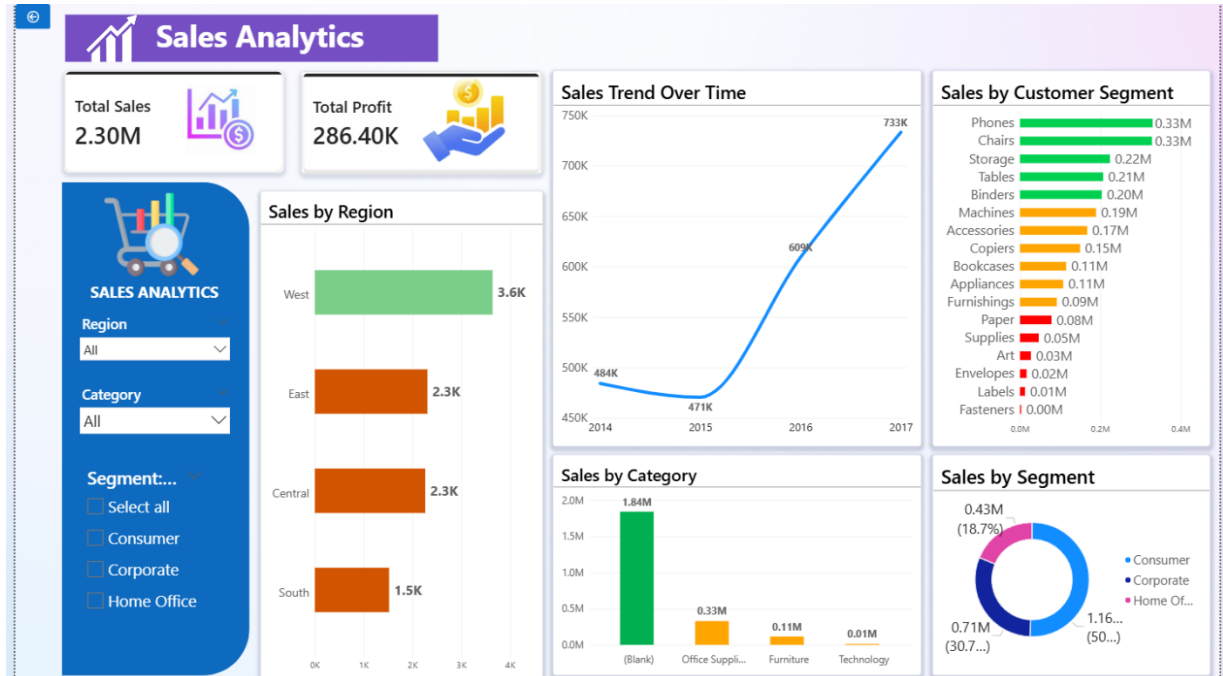
EXECUTIVE OVERVIEW



Explanation

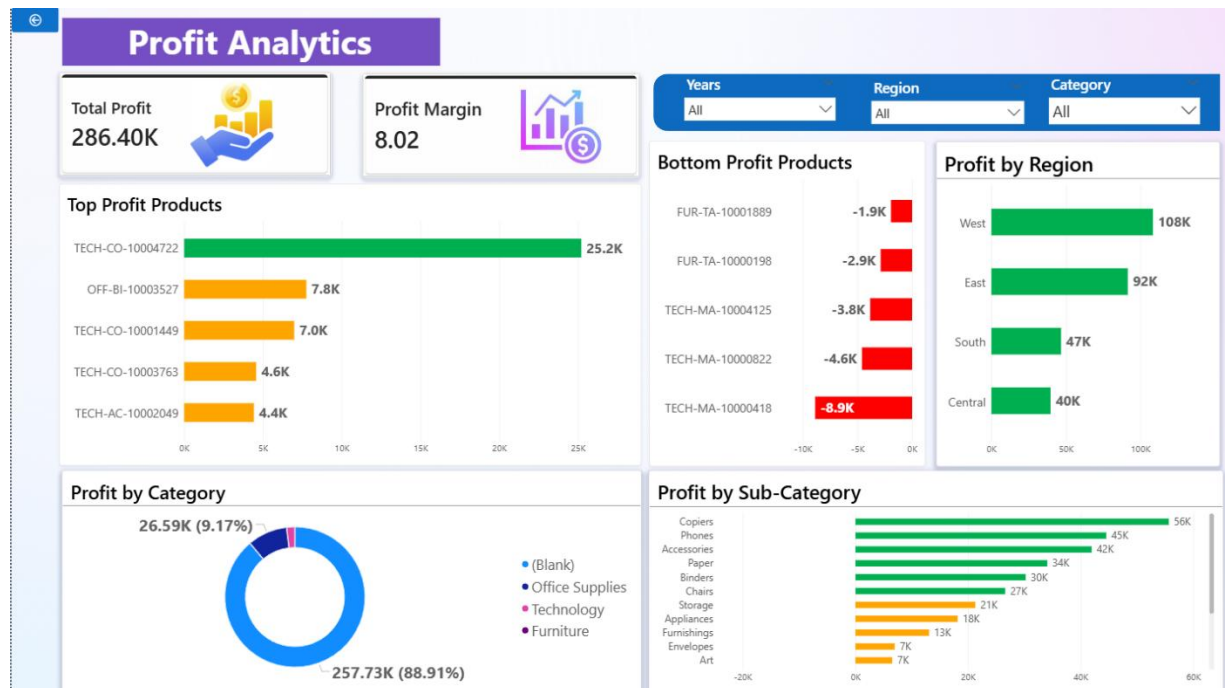
Analysis high-level business performance and executive metrics.

Sales Analytics



Explanation

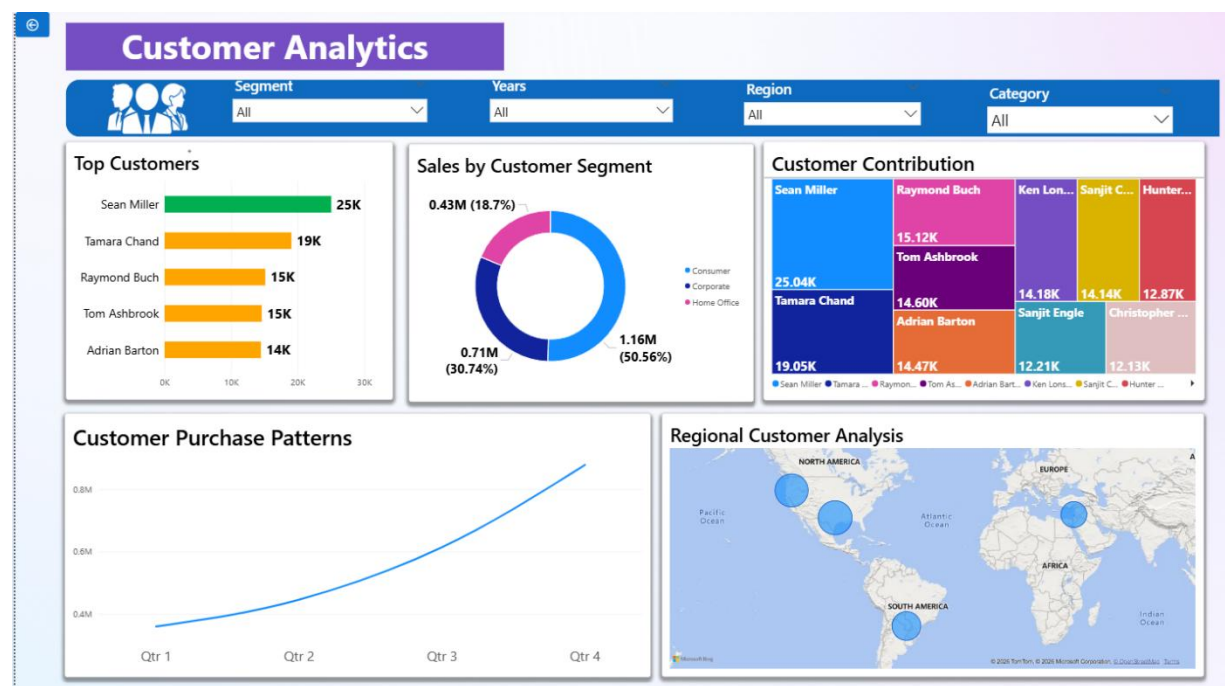
Analysis sales trends and revenue distribution. **Profit Analytics**



Explanation

Analysis profit margins and product profitability..

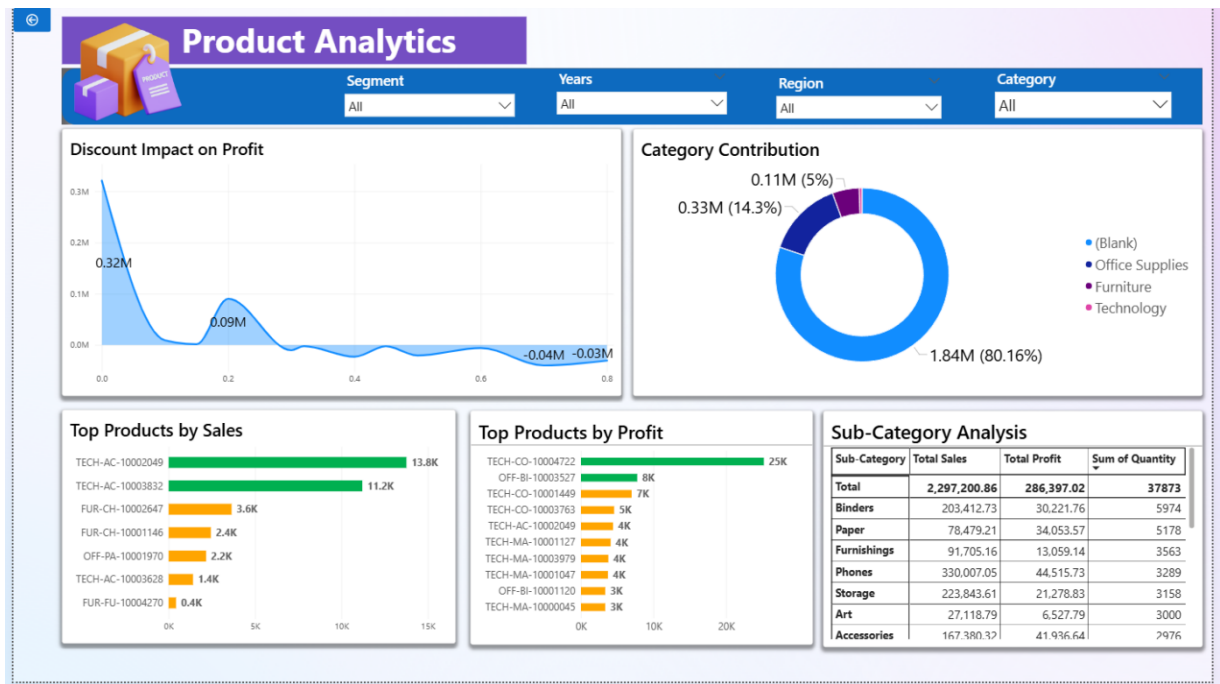
Customer Analytics



Explanation

Analysis customer behaviour and purchase patterns.

product Analytics



Explanation

Analysis product performance and discount impacts.

KEY INSIGHTS

INSIGHT-1

Sales volume spikes sharply in the final quarter of the year, peaking in November.

INSIGHT-2

Annual sales revenue shows continuous upward growth year-over-year since 2015.

INSIGHT-3

Technology products generate the highest overall net profit for the company.

INSIGHT-4

Customer purchase volumes increase steadily and consistently quarter-over-quarter.

INSIGHT-5

Offering higher discount rates drastically reduces and cuts into product profitability.

BUSINESS RECOMMENDATIONS

RECOMMENDATION-1

Focus more on the West Region to increase sales and profit.

RECOMMENDATION-2

Give special attention to Consumer customers as they generate the most sales.

RECOMMENDATION-3

Promote high-profit products like Copiers, Phones, and Accessories.

RECOMMENDATION-4

Reduce losses from Tables and Bookcases by improving pricing and sales strategies.

RECOMMENDATION-5

Keep valuable customers engaged through loyalty and reward programs.

FEATURES IMPLEMENTED

- ✓ Drill Down
- ✓ Drill Through
- ✓ Tooltips
- ✓ Dynamic Titles
- ✓ Slicers
- ✓ Conditional Formatting
- ✓ Navigation Buttons
- ✓ Custom Theme

CHALLENGES FACED

Data Cleaning Challenges Relationship
Management
DAX Optimization
Dashboard Layout Design

LEARNINGS

Power Query

Data Modeling

DAX Development

Dashboard Design

Storytelling

Business Analysis

PROJECT STORYTELLING

BUSINESS SCENARIO

Retail and e-commerce organizations generate large amounts of transactional data related to sales, profits, customer orders, product categories, and discount metrics.

However, decision-makers often struggle to convert this information into meaningful business insights that support revenue optimization, product management, and customer satisfaction.

BUSINESS PROBLEM

- Which business segments provide the highest revenue?
- How do monthly trends affect sales growth and order volume?
- Do high discount rates increase or hurt product profitability?
- Which customer groups demonstrate better purchasing behavior?
- What product categories influence overall net profit margins?

ANALYSIS APPROACH

- Analyzed business revenue across regional consumer segments.
- Compared sales growth against monthly and quarterly timelines.
- Evaluated product expenses, losses, and net profit margins.

- Measured the impact of heavy discounting strategies on earnings.
- Analyzed buyer satisfaction through repeat customer purchase patterns.

KEY FINDINGS

Year-end shopping spikes significantly increase monthly sales revenue. Technology products generate the highest overall net profits. Regular customer buying habits show steady quarterly growth. High product discount rates drastically reduce profit margins.

BUSINESS IMPACT

Supports revenue optimization planning.
Improves product portfolio management.
Enhances customer retention strategies.
Provides data-driven sales insights.
Enables better pricing decision-making.

PROJECT DEPLOYMENT

DEPLOYMENT OBJECTIVE

Deploy the Power BI retail solution to enable stakeholders to access interactive sales, profit, customer, and product dashboards in real time.

DEPLOYMENT PLATFORM

Power BI Service

DEPLOYMENT PROCESS

Power BI Service

STEP-1

Validate Retail Data Models and Sales/Profit DAX Measures in Power BI Desktop.

STEP-2

Optimize Dashboard Visuals, Slicer Interactivity, and Report Performance.

STEP-3

Publish the E-Commerce PBIX File to the designated Power BI Workspace

STEP-4

Configure Scheduled Dataset Refresh Settings for up-to-date transactional data.

STEP-5

Create Unified Executive Dashboards from Published Report tiles.

STEP-6

Apply Row-Level Security (RLS) for Regional Managers and Access Permissions.

STEP-7

Share the Interactive Dashboards with Retail Business Stakeholders.

BUSINESS BENEFITS

Centralized Retail Reporting Platform.

Interactive Sales and Profit Analytics.

Real-Time Transaction Performance Monitoring.

Improved Strategic Inventory Planning.

Faster Data-Driven Pricing Decisions.

CONCLUSION

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The Retail Sales & Financial Performance Analytics Dashboard successfully analyzed sales revenue, profit margins, customer behavior, product performance, and discount impacts.

The project generated valuable insights that can help organizations improve revenue optimization, product portfolio management, customer retention strategies, and business pricing decisions.